

Session 2026-06-18 — Public-Release Hardening, Distribution, and Scaling

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Reference doc for the work done the day ELI MKXI went public. Two threads ran together: **(A)** make the repo safe and clean for public distribution, and **(B)** make the whole surface — installer, launchers, docs, model path — consistent across **platform, hardware, and users**, scaled for the long term (**3B on a laptop → trillion-param on 8x datacenter GPUs**), *without sacrificing ELI's ethos* (100 % local, offline-by-default, no telemetry, model/user/hardware-agnostic, emergent persona).

1. Changelog (commits this session, on main)

Commit	Summary
a750bc3	ctx-tuner robustness + PII scrub + web UI. n_ctx_train-aware sizing on <i>both</i> load paths; _effective_ctx_limit = min(loaded n_ctx, n_ctx_train); prompt truncate-to-fit instead of crash; engine never surfaces raw GGUF streaming failed as a reply. Personal name/projects/research-IP genericized in code/tests/eval; personal .desktop untracked. FastAPI web chat UI at /.
bdfc9d9	Unified launcher + hardened server. scripts/eli_launch.sh (gui/serve/both); scripts/eli_serve.sh loopback-safe by default, --lan mints a token + prints the phone URL. Bearer-token gate on /v1/chat + /v1/execute (enforced only when LAN-exposed).
cd5ef44	Interactive installer. System report (CPU/RAM/GPU+VRAM/disk) → plan → confirm; auto-selects CPU/GPU build; offers a model sized to VRAM; modern summary. TTY-gated (piped/CI stays non-interactive). New flags --yes, --gpu, --auto-model/--model=/--no-model.
2a9a5a0	Remove hardcoded RTX 2060 references (comments/tests/dossiers).

Commit	Summary
88fee52	Fully hardware-agnostic — drop <i>all</i> specific GPU/CPU/box references (GeForce/3090/4090/A6000, “this box”, exact RAM). Backend identifiers (nvidia-smi/CUDA/Metal/MPS) kept — those are the tech stack, not a machine.
(this commit)	Scaling + consistency pass — multi-GPU installer report, README scaling section, this blueprint.

Earlier in the session (pre-public, not all committed)

- **Context tuner** root-cause fix: the adaptive fallback loader ignored `n_ctx_train` and loaded a 4096-train model at 16384 (“training context overflow”) → garbled output and GGUF streaming failed: Requested tokens exceed context window surfaced as a reply. Fixed at the loader, the sizing math, the overflow retry, and the engine output path.
- **GUI smart-fit** floors `ctx` at ELI’s brief+gen need so a capable model is never chopped.
- **LoRA pipeline** reorg → training/ (extract/train/merge for Qwen3-8B), --from-db voice extractor (2.5k replies from conversation_turns, state excluded), bad-pattern filter (drops error-leakage/confab/fragment replies). (*gitignored, local.*)
- **Local data wipe**: DBs/FAISS/conversations cleared, identity reset to empty → clean first-run instance. Backup: ~/eli_personal_backup_20260618_135332.tar.gz.

Later in the session (post-scaling — release polish + correctness)

- **Full DB architecture on fresh install**: new `eli/core/init_data.py` builds every store + table up front (user/agent/system_index/coding_memory) — schema only, **zero personal rows**; wired into both installers. `agent_bus.ensure_agent_tables()` added.
- **User Model split-brain fixed**: `profile_extractor._user_db()` hardcoded <repo>/artifacts/db while the reader used `paths.user_db_path()` → on an installed package the model was written and read from *different* files. Now both use the canonical path (round-trip verified).
- **ctx sizing**: VRAM reserve default **1500 → 250 MB**; `ELI_CTX_FRACTION` GUI default **0.65 → 0.9** (matched to the loader) so launch `ctx` isn’t needlessly capped.
- **Model catalog expanded to 7** (was 3): qwen2.5-3b, qwen2.5-7b (default), qwen3-8b, falcon3-10b, phi-4 (MIT), qwen3.6-35b-a3b (Apache-2.0), falcon-h1-34b. All URLs verified 200, licences checked. **Required nomic embedder auto-installs**; vision is optional aux. **Multi-select download** (--choose) wired into both installers (pick any number, or none).
- **Habits**: smarter detection (recurrence across ≥3 distinct days, not a same-day burst); `CONFIRM_HABIT` runs the habit immediately if today’s slot already passed.
- **Onboarding contamination fixed**: the interview consumed every mid-interview message as an answer, so user questions were stored as `identity.role / preference.style`. Now a question/command bows out instead of being captured; contaminated rows purged.
- **Test isolation fixed (was injecting false memories)**: `confstest` set `ELI_ARTIFACTS_DIR` but the user DB resolves via `ELI_USER_DB/ELI_MEMORY_DB`, so `store_memory()`/reflection writers polluted the real user.sqlite3. Pinned via `ELI_USER_DB` → no test can touch the real DB (full suite: real DB held at 10).
- **De-personalization (multi-user)**: physics-framework bias ($\exists/\chi/\phi$, stueckelberg/lagrangian/scalar/fenics/openfoam/meep/tokamak) removed from 8 runtime files → generic science detection.
- **GUI declutter**: Orchestration + Test & Review **demoted to Labs sub-tabs**; Background-job listing unified into the **Tasks** tab (Coding tab’s duplicate removed). 12 main tabs.
- **Docs**: capabilities doc regenerated (**183/183** coverage); blueprints/ELI_USER_MANUAL.md (+ PDF, with flowcharts + cognition/memory appendices); finetuning_guide.md made model-choice-first + hardware-agnostic; capabilities PDF.
- **CI**: cross-platform smoke (ubuntu/macos/windows × py3.10/3.12) green throughout.

2. Consistency audit (platform · hardware · users)

[yes] Consistent / verified

- **Public repo carries no personal data.** artifacts/ (DBs, conversations, FAISS, memories), blueprints/, training/ all gitignored — 0 tracked. Verified autobuild: a fresh clone creates an empty 28-table user.sqlite3 with **0 data rows**.
- **No machine/model/user hardcoded** anywhere in tracked source (swept twice).
- **Platforms covered:** Linux/CUDA, Windows (install.bat→install.ps1), macOS/Metal (install.sh Darwin branch + requirements-macos.txt), Android/Termux (headless, expert-only — honestly scoped).
- **Offline-by-default** holds: netguard fail-closes outbound; the inbound web server does not breach it (binding/listening ≠ connecting out).

[!] Findings — fixed this session

- **Installer showed 1 GPU** (nvidia-smi ... | head -1) → now enumerates all GPUs and reports **total VRAM** (N× <name> (... MiB total)).
- **Phone server bound 0.0.0.0 with no auth** → now loopback-safe by default; LAN mode requires a token (gates the system-action endpoint too).
- **README model recommendation drift** (Qwen2.5-7B vs the session's Qwen3-8B) → README now documents the scaling path + --auto; Qwen3-8B is the documented LoRA/quality target.

[!] Findings — documented, not yet fixed (roadmap, §4)

- **install.ps1 (Windows) is not yet at parity** with the new interactive install.sh (system report / plan / model offer). Functional, but plainer. *Parity port pending*.
- **requirements pin drift:** requirements.txt pins llama_cpp_python==0.3.20, requirements-full.txt uses >=0.3. Harmonize to one policy (frozen lock is canonical).
- **Capability count:** README says “208 capabilities”; runtime manifest reports 208. Make the README pull the live number or reconcile.

3. Scaling — 3B → trillion-param (the long-term contract)

ELI is **agnostic by construction**; the *same install* must span a Raspberry-Pi-class box to a \$250k 8× datacenter-GPU server. How each axis scales today:

Axis	Small end	Large end	Mechanism
Context	4k models warned + truncated gracefully	256k-train models capped to their real n_ctx_train	GGUF meta-data read pre-load; min(loaded, train) ceiling
VRAM fit	shed layers→batch→ctx to fit 4 GB	uses summed VRAM across all GPUs	hardware_profile sums all GPUs; smart-fit per machine
Model size	3B–35B built-in catalog (7 models)	any GGUF via drop-in or catalog override	models/*.gguf scan + ELI_MODEL_CATALOG

Axis	Small end	Large end	Mechanism
Quantization	Q4/Q5 on tiny cards (KV-quant <12 GB)	full precision when VRAM allows	tier- derived cache_type_k/v

The model catalog is intentionally a starter set (1.5B / 3B / 7B with verified URLs). Scaling beyond it is the **override mechanism** (already implemented, no code change): drop a .gguf in models/, or supply a catalog.json (same schema) for mid/large/huge models. This keeps the built-in download honest (no unverifiable large-model URLs baked in) while making any scale reachable.

Ethos at every tier (non-negotiable)

Local-first does **not** weaken as hardware grows — an 8×GPU server is *on-prem*, not cloud. Offline-by-default, no telemetry, emergent persona, and your-data-on-your-hardware hold identically from 3B to trillion. Scaling up is a hardware fact, never an ethos compromise.

4. Roadmap / known limits (for long-term maintainability)

1. **Multi-GPU tensor-split for a single huge model.** `select_gpu()` currently picks the single GPU with the most free VRAM; a trillion-param model needs llama.cpp `tensor_split` / device mapping across all GPUs. VRAM *totals* are already summed — wire the loader to pass `tensor_split/main_gpu` when $N > 1$ GPUs are present.
2. **install.ps1 interactive parity** with `install.sh` (report → plan → model offer).
3. **Requirements harmonization** — single pin policy across `requirements*.txt`.
4. **GUI first-run welcome wizard** — guided *welcome* → *detected hardware* → *recommended model* → *one-click download (progress)* → *name* → *done*, replacing the “open Settings” dead-end for zero-model users. (Installer side is now interactive; GUI side pending.)
5. **Catalog tiers** — once verified, add Qwen3 (8B/14B/32B) + a large tier to the built-in catalog so `--auto` reaches higher without a manual `catalog.json`.
6. **Commit-history email** — older commits carry a personal gmail; current ones use the GitHub noreply. A `git filter-repo` rewrite would scrub history (heavy; owner’s call).

5. Operational notes

- Launch: `./scripts/eli_launch.sh (desktop) · ... serve --lan (phone web app) · ... both`.
- Server is loopback+tokenless by default; `--lan` exposes to the network *with* a token.
- Personal backup (restore the wiped local instance): the tarball in `$HOME`.
- blueprints/ and training/ remain gitignored (local-only reference + tooling).